

CSCE 190: Computing in the Modern World

Class 6 – Problem Solving with Computing

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Carolinian Creed: “I will practice personal and academic integrity.”

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Outline

- Introduction Section
 - Recap from Last Class
- Main Section
 - Problem Solving Process
 - Validating the Problem
 - Selecting Problems that Matter
- Conclusion Section
 - Ask Me Anything

Introduction Section

Recap from Last Lecture - Class 5

We talked about

- Agentic AI and Safety
- Web Sites, Web Pages
- GitHub Pages – example of sharing, HTML/CSS and Markdown

Assignment 5 – Github Website

- Create a Github account
- Create a github.io repository on Github
- Connect to your repo and Commit a change
- Using VS Code, the Github website, Github dev, etc.
- Test your github.io link

Recent Discussion – AI and Safety

“Researchers have long warned that three conditions could lead to loss of control:

- a misaligned goal,
- the capability to pursue it, and
- an environment that allows it,”

ACM A.M. Turing Award laureate Yoshua Bengio said at UN
(Sep 2026)

<https://thehill.com/policy/technology/6102955-un-ai-panel-urges-safeguards/>



Computing in the News

- The Vanishing Apprentice, Sep 2026, <https://cacm.acm.org/news/the-vanishing-apprentice/>
 - “The first prototype was always the easy part,” he said. “**Roughly 95% of the cost of a software product occurs after the initial version is released.** Turning a prototype into a stable, reliable, and evolvable system is where the real work starts.”
 - Lo believes the larger risk is not just a decline in entry-level jobs, but **an eventual shortage of experienced engineers capable of maintaining the vast codebases now being generated with AI tools.**
 - [No one wants to deal with AI slop]
- OpenAI hacking, Sep 2026, <https://www.nytimes.com/2026/09/25/technology/openai-ai-us-government-websites.html>
 - ... meddled with the **websites for the Education Department, the Commerce Department** and the Securities and Exchange Commission this summer without the A.I. lab’s knowledge, according to security researchers and a person familiar with the episodes.
 - [Hacking by another name ... out-of-control car would be dealt similarly ?]

Main Section

Problems

- Start with something that we needs to be fixed
- Answer
 - For whom is the problem important?
 - What happens if you solve it well?

Problems and Their Statements

Good problem statements:

- Are focused on the user (avoid using “we” or “the product”)
- Are broad (leave room for innovation and alternatives)
- Are manageable (just this one problem)

Source: Robert Bailey, CSCE 190 Instructor

Problem Solving with Computing

- Problem Solving Process
- Validating the Problem
- Selecting Problems that Matter

Problem Solving Process

Receiving the problem *{from client}*

- Client creates **requirements** document
- **Understanding the problem and scoping**
 - Developer creates **specification**
- **Solving**
 - Understand and implementing with tools for solving the problem: **Design docs, test plans**
 - Create solutions: **Code, Release**
 - Evaluate solutions: **Test logs**
- **Delivering *{to customers}***

A generic process involves:

- Customer:
 - requirements
- Developer:
 - specification
 - design
 - development (coding)
 - testing
 - release
- Customer:
 - evaluation
- acceptance

The above process is highly simplified and rarely happens in one-round. Read about 'Software Development Life Cycles' (SDLCs).

Validating the Problem

- **Objective:** Understanding the problem and scoping
- Think and articulate – what will be
 - The inputs: e.g. data, current process
 - The outputs (deliverables): e.g., software, solution, documentation
 - Criteria for problem being solved: e.g., metrics
- Considerations
 - Current practices/ processes
 - Why would someone use your solution to the problem?
 - **If you can't identify inputs, outputs, and evaluation metrics, the problem is most likely ill-defined computationally. Needs more work!**

SMART Metrics

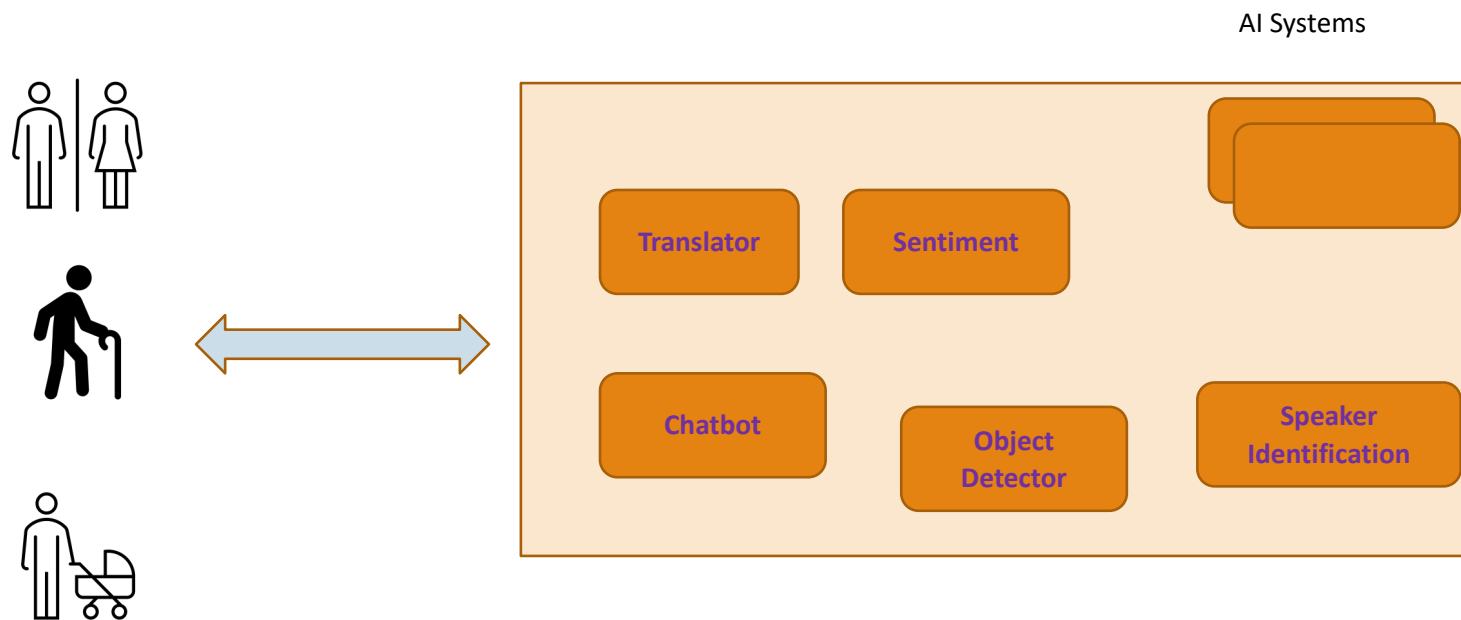
- ***Specific***: Targeting a particular area for improvement
- ***Measurable***: Quantifying, or at least suggesting, an indicator of progress
- ***Assignable***: Defining responsibility clearly
- ***Realistic***: Outlining attainable results with available resources
- ***Time-related***: Including a timeline for expected results

Source: https://en.wikipedia.org/wiki/SMART_criteria

Selecting Problems that Matter

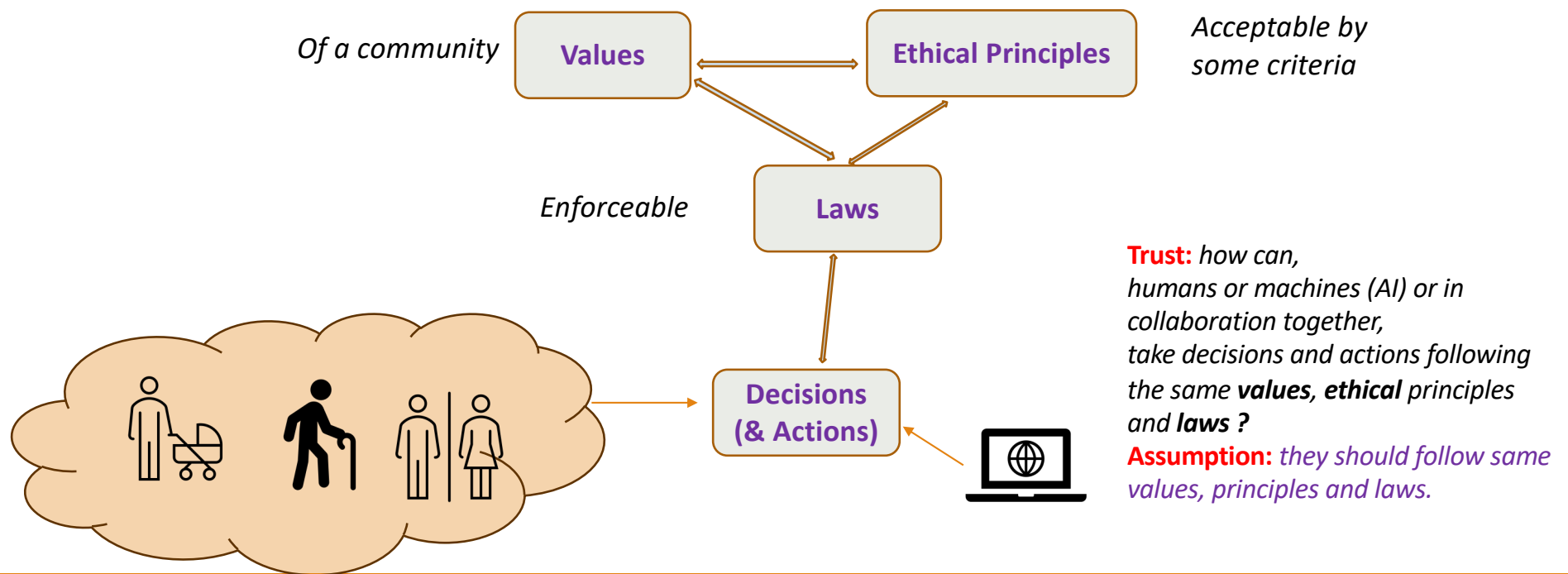
- Answer
 - For whom is the problem important? **[A lot of people]**
 - What happens if you solve it well? **[With change the status quo for the better]**
 - Do you have the inputs to get started? **[Yes? Otherwise, re-think problem or inputs]**
 - Do you have the resources/ are you best qualified to solve the problem? Do you need to augment the team? **[Yes? Otherwise, pass or get help]**

Technology and People



Trust: Can people trust *new system* to perform capably, consistently, and with human values?

Values, Ethics, Law, Decisions (Actions) with Trust



Illustrative Problem Category:

1. Helping students
 1. With class commitments
 2. With meals
 3. With finances (?)
 4. ...

Problem Example

- **Problem:** build a system to help students choose meals during study days
- Validating the problem:
 - The inputs: e.g. data, current process
 - The outputs (deliverables): e.g., software (app)
 - Criteria for problem being solved: e.g., adoption rate
- Solving the problem
 - Recommendation methods
- Trustworthiness ecosystem
 - Values, ethical principles, laws
 - Users impacted

Problem statement criteria

- Are focused on the user (avoid using “we” or “the product”)
- Are broad (leave room for innovation and alternatives)
- Are manageable (just this one problem)

Assignment 6

- Group 4-5 students
- Discuss a problem category
- Decide on a problem – see *problems that matter*
- Write a problem statement
 - The group will have one problem statement
 - Add your statement to your GitHub in a sub-folder called 'assignment-6'
 - Do mention all the students in your group

Concluding Section

Conclusion

We talked about

- Problem Solving Process
- Validating the Problem
- Selecting Problems that Matter

Week	Content	Assignment
1	Intro, syllabus, university resources	Slack account
2	CSE majors, career paths, research areas	Write a mission statement
3	Github (account and basics) + Lab	Github account, repo
4	HTML basics, IDE (VS Code)	None
5	More HTML, CSS, Problem Solving	Update Github with CSS
6	Brainstorming, problem statement	Problem statement
7	Computing – (Software) Dev. Life Cycle	Draft Requirement, Specification
8	Computing – Testing a System	Test Plan
9	AI – LLMs/FMs, Generative AI, Issues and Challenges	Problem Solving with GenAI
10	Building a resume, LinkedIn	Draft resume in PDF
11	Review resumes	Review resumes, update Github
12	Convert resume to HTML	Begin conversion, update Github
13	CSE research areas	(Finalize website)
N/A	Last day of classes: 4 Dec, 2026	All extensions end

Ask Me Anything
